INORGANIC CUMULATIVE EXAM

April 25, 1992

The following figures were removed from Shriver, Atkins and Langford, Inorganic Chemistry, chapter 14, Bonding and Spectra of Complexes. Some of the labels have been marked over. Indicate in your exam blue book what the labels should be and give a brief explanation of each. Your explanation should point out the importance of the label in the diagram.

Please key your answers to the figure numbers.

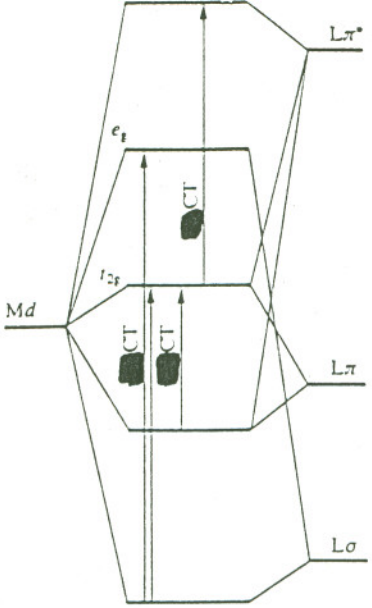

Fig. 14.7 A summary of the charge-transfer transitions in an octahedral complex.

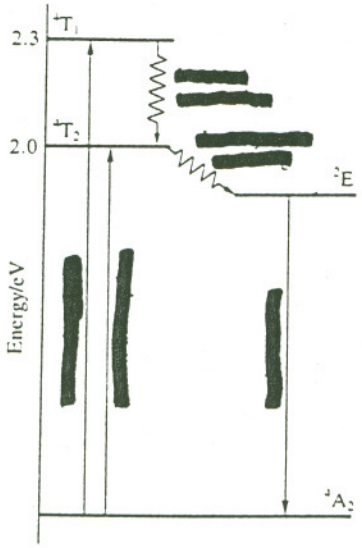

Fig. 14.8 The transitions responsible for the absorption and luminescence of Cr^{3+} ions in ruby.

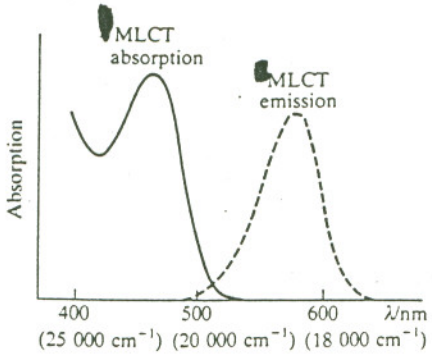

Fig. 14.9 The absorption and phosphorescence spectra of $[Ru(bipy)_3]^{2+}$.

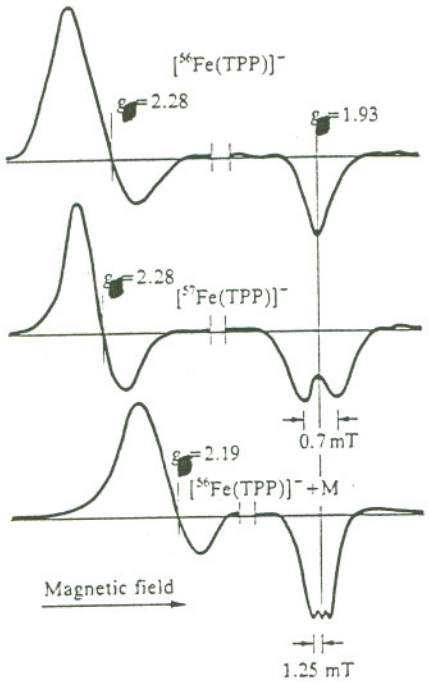

Fig. 14.22 The EPR spectra of the anion produced by reduction of the square-planar iron tetraphenylporphyrin $[Fe(TPP)]^-$, of labeled $[^{57}Fe(TPP)]^-$, and of $[Fe(TPP)]^-$ in the presence of excess pyridine.

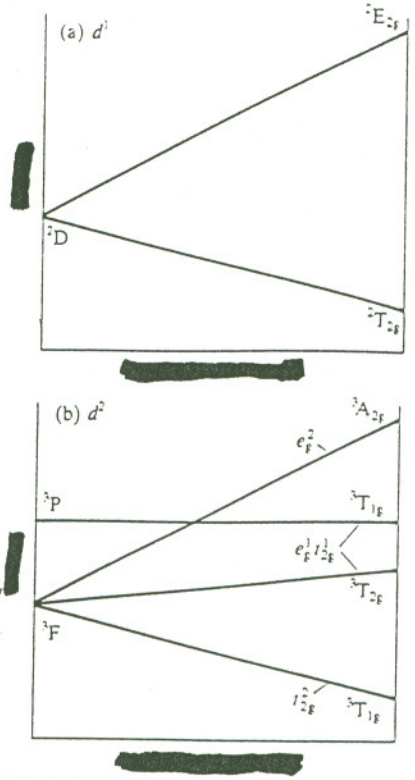

Fig. 14.2 Correlation diagram between a free ion and the strong-field terms of (a) a d^1 configuration and (b) a d^2 configuration.

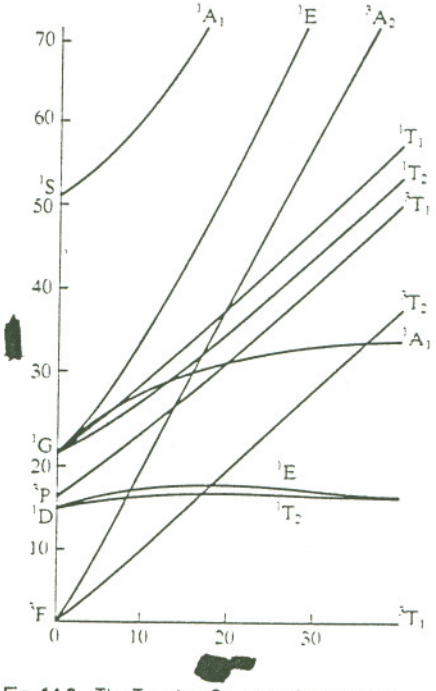
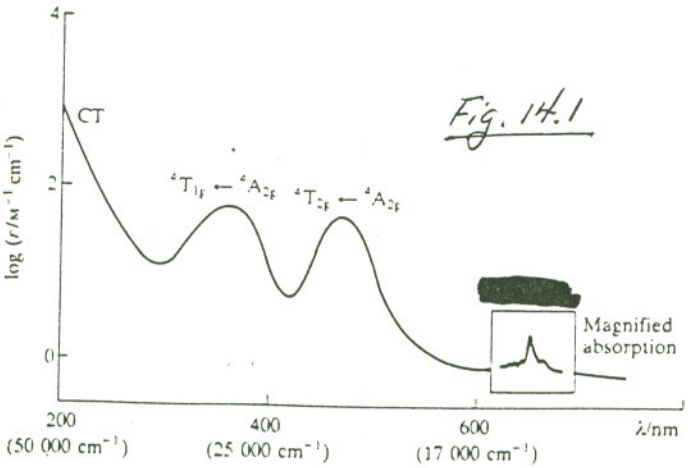
Fig. 14.3 The Tanabe–Sugano diagram for the d^2 configuration. A complete collection of diagrams for d^n configurations is given in Appendix 7 at the back of the book.